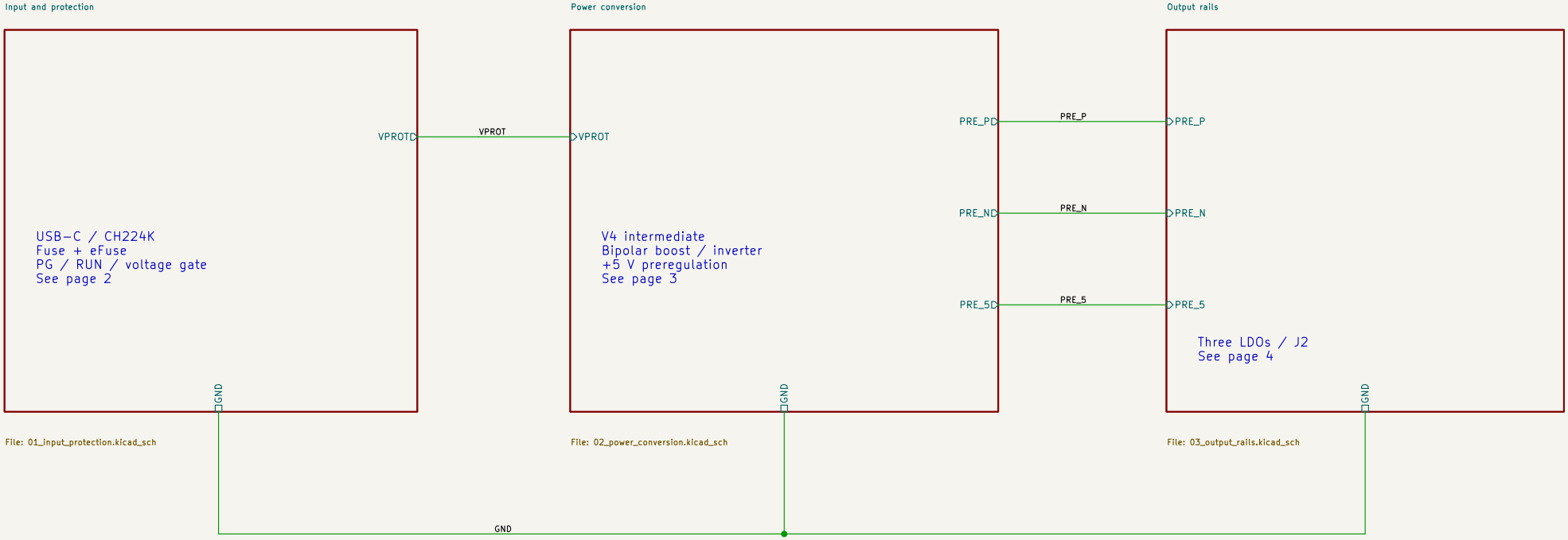
BIG MODULAR / SCHEMATIC MAP

Three functional subsheets | 88 real components | Circuit connectivity retained

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

LIMITS: No PDO/current proof, qualified input surge protection or completed PCB layout.
Read docs/schematic-guide.md and docs/component-guide.md.

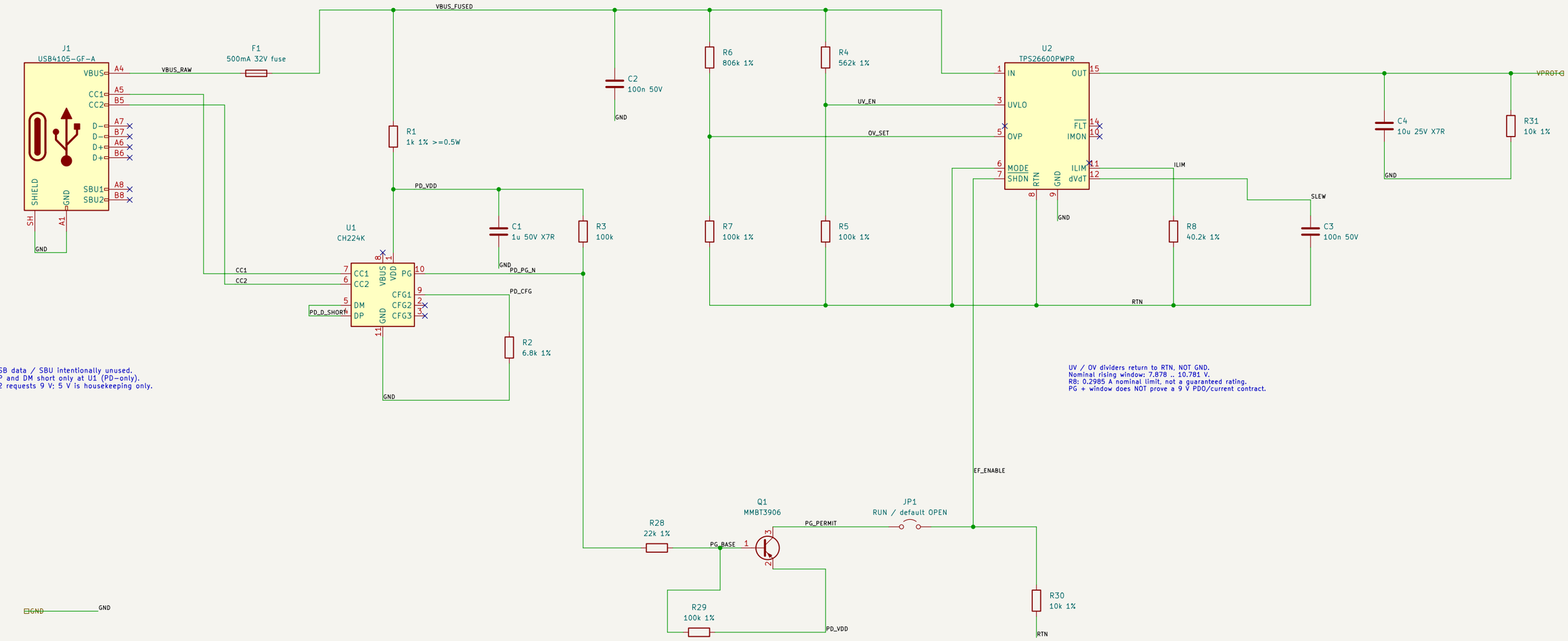


HOW TO READ
Open each sheet in KiCad or click its box in the PDF.
Local feedback, filtering and protection networks use real wires.
Named ports carry power between sheets; GND is shared, RTN is not.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE
Big Modular
Sheet: /
File: big_modular.kicad_sch
Title: Big Modular — Overview and sheet navigation
Size: A4 | Date: | **Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6 | Id: 1/4

01 / INPUT AND PROTECTION

WIP — DO NOT FABRICATE | Read docs/schematic-guide.md | Return to overview using the sheet navigator



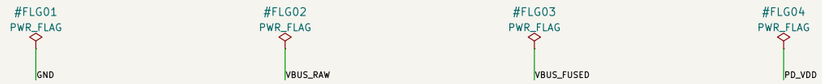
USB data / SBU intentionally unused.
DP and DM short only at U1 (PD-only).
R2 requests 9 V; 5 V is housekeeping only.

UV / OV dividers return to RTN, NOT GND.
Nominal rising window: 7.878 .. 10.781 V.
R8: 0.2985 A nominal limit, not a guaranteed rating.
PG + window does NOT prove a 9 V PDO/current contract.

No qualified VBUS / CC ESD or surge protection. D1 is absent.
All converter bulk is downstream of U2. See design-review.md.

PG low -> Q1 on -> series RUN -> SHDN high.
Default RUN OPEN: R30 defaults SHDN low.
R31 bleeds VPROT; zero residual voltage is NOT guaranteed.

ERC source declarations (not physical parts)



INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Input and protection/
File: 01_input_protection.kicad_sch

Title: Big Modular — Input and protection

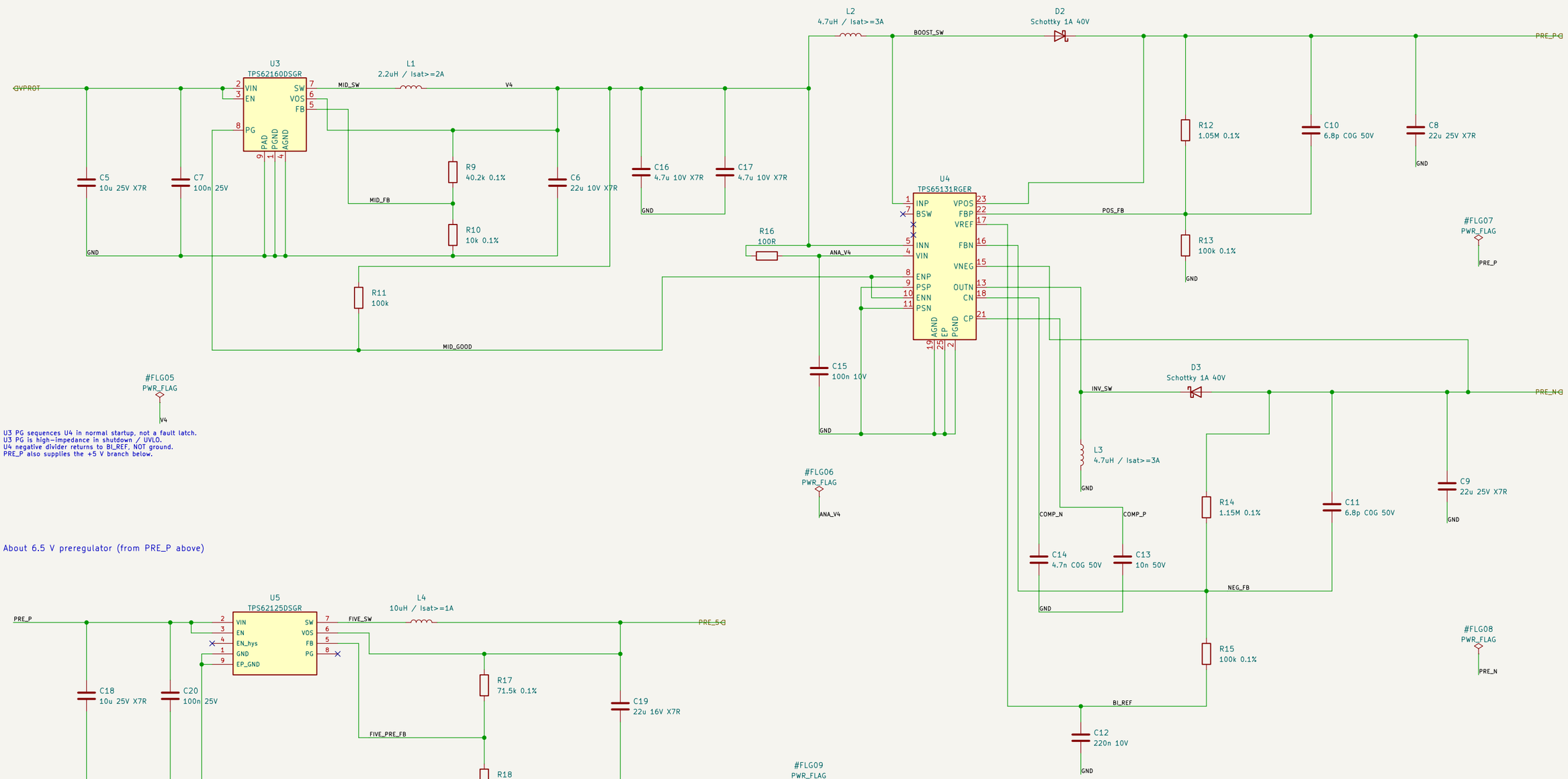
Size: A3	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 2/4

02 / POWER CONVERSION

WIP — DO NOT FABRICATE | Read docs/schematic-guide.md | Return to overview using the sheet navigator

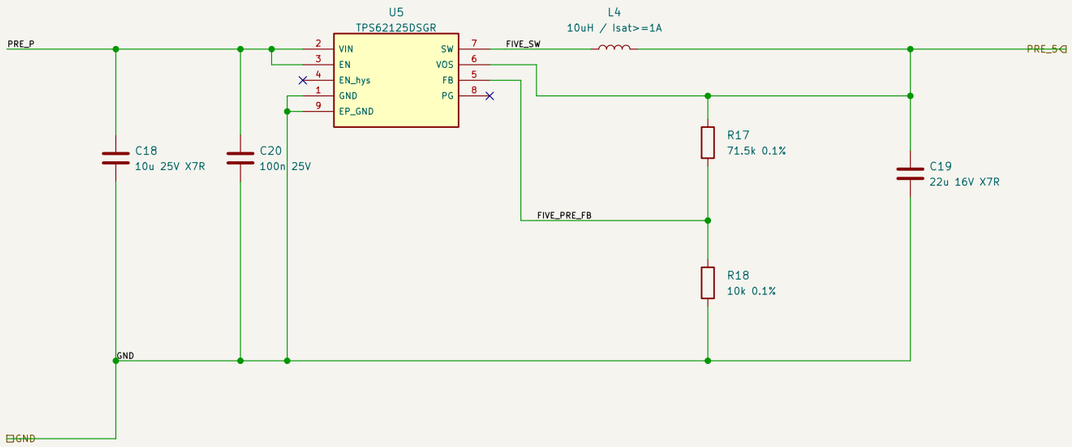
4.016 V intermediate

Bipolar preregulator: about +/−13.95 V



U3 PG sequences U4 in normal startup, not a fault latch.
U3 PG is high-impedance in shutdown / UVLO.
U4 negative divider returns to BI_REF, NOT ground.
PRE_P also supplies the +5 V branch below.

About 6.5 V preregulator (from PRE_P above)



INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Power conversion/
File: 02_power_conversion.kicad_sch

Title: Big Modular — Power conversion

Size: A3	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 3/4

03 / OUTPUT RAILS

WIP — DO NOT FABRICATE | Read docs/schematic-guide.md | Return to overview using the sheet navigator

+12 V / low-noise LDO

D4
Schottky 1A 40V

U6
TPS7A4901DGNR
IN
EN
NR/SS
NC
DNC
OUT
FB
GND
EP_GND

~~PRE_P~~
P12_NR
P12_FB
P12

C21
10u 25V X7R
C23
10n 50V
R19
91.2k 0.1%
C24
10n 50V
C22
10u 25V X7R
D5
Schottky 1A 40V
R25
100k
R20
10k 0.1%

GND

−12 V / negative LDO

D6
Schottky 1A 40V

U7
TPS7A3001DGNR
IN
EN
NR/SS
NC
DNC
OUT
FB
GND
EP_GND

~~PRE_N~~
N12_NR
N12_FB
N12

C25
10u 25V X7R
C27
10n 50V
R21
92k 0.1%
C28
10n 50V
C26
10u 25V X7R
D7
Schottky 1A 40V
R26
100k
R22
10k 0.1%

GND

+5 V / low-noise LDO

D8
Schottky 1A 40V

U8
TPS7A4901DGNR
IN
EN
NR/SS
NC
DNC
OUT
FB
GND
EP_GND

~~PRE_5~~
P5_NR
P5_FB
P5

C29
10u 25V X7R
C31
10n 50V
R23
32.2k 0.1%
C32
10n 50V
C30
10u 25V X7R
D9
Schottky 1A 40V
R27
47k
R24
10k 0.1%

GND

~~GND~~

J2 OFF-BOARD
16-pin POWER OUT
1/2: −12 V / red stripe
3..8: ground
9/10: +12 V
11/12: +5 V
13..16: NC

J2
16-pin IDC POWER OUT
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
N12
P12
P5
GND

Negative LDO: IN / EN use PRE_N.
GND and EP remain at 0 V.
Diodes address reverse differential / polarity,
NOT precision overvoltage protection.

10uF MLCCs: verify capacitance under bias.
No rail-current rating, tracking or rapid-discharge guarantee.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Output rails/
File: 03_output_rails.kicad_sch

Title: Big Modular — Output rails

Size: A3	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 4/4